

Uniform Sampling of k -Proper Colorings in Graphs

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Problem Statement: Sampling Proper Colorings

What is a k -proper coloring?

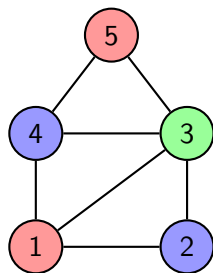
Given a graph $G = (V, E)$, a coloring $\sigma : V \rightarrow \{1, \dots, k\}$ is **proper** if no two adjacent vertices share the same color:

$$\sigma(u) \neq \sigma(v) \quad \forall (u, v) \in E.$$

The Sampling Goal

Let Ω be the state space of all valid k -proper colorings of G . Our objective is to draw a coloring σ uniformly at random from this space [2]:

$$\sigma \sim \pi, \quad \text{where} \quad \pi(\sigma) = \frac{1}{|\Omega|}.$$



Example: A valid 3-coloring of a graph. Note that the chord $(1, 3)$ forces vertices 1 and 3 to be different colors.

Why is Sampling Hard?

The Counting Bottleneck

The total number of valid colorings is given by the graph's chromatic polynomial, $|\Omega| = P_G(k)$.

Evaluating $P_G(k)$ is identically **#P-complete** [2].

$\implies$ *Consequence:* Exact counting is computationally intractable for general graphs, which rules out naive combinatorial methods for uniform sampling.

The Workaround: Markov Chain Monte Carlo (MCMC)

Instead of sampling exactly, we generate an *approximate* uniform sample by simulating a random walk:

- 1 **Construct** an ergodic Markov chain where the state space is Ω .
- 2 **Design** the transition probabilities so the stationary distribution π is exactly uniform.
- 3 **Simulate** the chain until it approaches equilibrium.

Standard MCMC Algorithm

Update Rule [1]

Given a graph with degree at most Δ , to obtain the coloring X_t from X_{t-1} :

- ❶ **Select** a vertex v uniformly at random from the vertices.
- ❷ **Sample** a color c uniformly at random from the k colors.
- ❸ **Check Validity:** If coloring vertex v with color c results in a valid, proper coloring, set $X_t(v) = c$.
- ❹ **Reject:** If it would not result in a proper coloring, reject the update and set $X_t = X_{t-1}$.

Intuition: We only ever modify one vertex at a time. The check prevents us from ever leaving the state space of strictly proper colorings.

Convergence: Why Does This Work?

To guarantee our algorithm converges to a target distribution, we rely on the **Fundamental Theorem of Markov Chains**. Our state space Ω is finite, and the chain satisfies two key properties:

- **Aperiodic:** A state can transition to itself. Because the algorithm rejects invalid colors, $P(\sigma, \sigma) > 0$ (self-loops exist).
- **Irreducible:** Any valid coloring can transition to any other valid coloring in finite steps. This is guaranteed provided $k \geq \Delta + 2$, ensuring the state space Ω is connected [2, 1].

Fundamental Theorem of Markov Chains

If a finite Markov chain is irreducible and aperiodic (i.e., ergodic), it possesses a **unique stationary distribution** π to which it converges from any initial starting state.

Targeting the Uniform Distribution

Theorem

The unique stationary distribution of the Metropolis dynamics over proper colorings is uniform:

$$\pi(\sigma) = \frac{1}{|\Omega|}.$$

Proof Sketch [1].

Consider two proper colorings σ and τ that differ *only* at vertex v . For the Metropolis chain, the probability of transitioning from σ to τ is:

$$P(\sigma, \tau) = \frac{1}{n} \times \frac{1}{k}$$

By symmetry, the probability of transitioning back from τ to σ is exactly the same:

$$P(\tau, \sigma) = \frac{1}{n} \times \frac{1}{k}$$

Since $P(\sigma, \tau) = P(\tau, \sigma)$, the transition matrix is symmetric. For an irreducible, aperiodic Markov chain, symmetric transition probabilities guarantee that the stationary distribution is precisely uniform. □

The Next Question: How Fast?

We know the chain eventually reaches the uniform distribution. **But how long does it take?** To measure this, we first need a notion of distance between our current distribution at time t and the target distribution π .

Definition (Total Variation Distance [1])

The total variation (TV) distance between two distributions μ and ν on Ω is:

$$\|\mu - \nu\|_{TV} = \frac{1}{2} \sum_{\sigma \in \Omega} |\mu(\sigma) - \nu(\sigma)| = \max_{A \subseteq \Omega} |\mu(A) - \nu(A)|$$

Definition (Mixing Time [1])

The mixing time is the number of steps required to get close to the stationary distribution, starting from the worst-case initial state x :

$$\tau_{mix} = \max_{x \in \Omega} \min \left\{ t \geq 0 : \|P^t(x, \cdot) - \pi\|_{TV} \leq \frac{1}{2e} \right\}$$

Standard Mixing Time Result

Theorem

Let $G = (V, E)$ be a graph with maximum degree Δ . The Metropolis Markov chain over proper colorings will mix in time polynomial in the size of the graph, provided $k \geq 4\Delta + 1$. Specifically, the mixing time is bounded by:

$$\tau_{mix} \leq kn(2 + \log n)$$

Our Goal

We will prove this mixing bound using a coupling argument. We will define a joint process (X_t, Y_t) and track the expected number of disagreeing vertices.

What is a Coupling?

Definition (Coupling of Distributions [2])

A coupling of two probability distributions μ and ν on a state space Ω is a joint distribution (X, Y) on $\Omega \times \Omega$ such that its marginals satisfy:

$$X \sim \mu \quad \text{and} \quad Y \sim \nu$$

Coupling for Markov Chains [1]

For a Markov process, a coupling is a joint stochastic process (X_t, Y_t) such that:

- 1 Individually, X_t and Y_t both follow the exact transition rules of the Markov process.
- 2 They are allowed to be correlated (i.e., they can look at each other's random coin flips).
- 3 Once they meet ($X_t = Y_t$), they move together forever ($X_{t+1} = Y_{t+1}$).

The Coupling Lemma

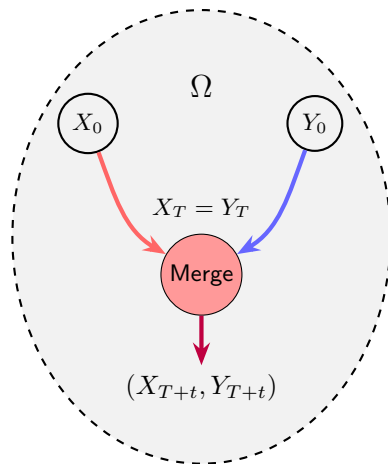
We need to bound the TV distance without calculating the giant state space matrix P .

Lemma (Coupling Lemma [1])

Let (X_t, Y_t) be a valid coupling starting from states s and s' .
Then:

$$\|P_s^t - P_{s'}^t\|_{TV} \leq \mathbb{P}(X_t \neq Y_t \mid X_0 = s, Y_0 = s')$$

Our Strategy: Construct a "smart" coupling and prove they are highly likely to have matched ($X_t = Y_t$) within a polynomial number of steps.



The Coupling Construction: Progress

Let Z_t denote the number of vertices in the graph for which X_t and Y_t assign different colors.

The Coupling Rule [1]: At each step, Y_t chooses the exactly same node v and the same color c as X_t .

Good Case: Progress ($Z_{t+1} = Z_t - 1$)

- We must pick one of the Z_t vertices for which the colors differ.
- We must pick a color c that is not one of the $\leq \Delta$ colors of a neighboring vertex in X_t , nor in Y_t .
- There are at least $Z_t(k - 2\Delta)$ such good choices of v and c .

Probability of Progress

$$\mathbb{P}(\text{Progress}) \geq \frac{Z_t(k - 2\Delta)}{nk}$$

The Coupling Construction: Setback

Recall: Z_t is the number of differing vertices, and Y_t copies the update choices (v, c) of X_t .

Bad Case: Setback ($Z_{t+1} = Z_t + 1$)

- We must pick a vertex that is a neighbor of one of the Z_t vertices on which the colorings differ.
- We must pick a color c which is the color of that neighbor in one of the two chains.
- There are at most $2\Delta Z_t$ such bad options.

Probability of Setback

$$\mathbb{P}(\text{Setback}) \leq \frac{2\Delta Z_t}{nk}$$

Putting It Together: The Contraction

We calculate the expected change in the number of differing vertices, Z_t [1]:

$$\mathbb{E}[Z_{t+1} \mid Z_t] \leq Z_t + \underbrace{\frac{2\Delta Z_t}{kn}}_{\text{Bad}} - \underbrace{\frac{Z_t(k - 2\Delta)}{nk}}_{\text{Good}}$$

Simplifying this expression yields:

$$\mathbb{E}[Z_{t+1} \mid Z_t] \leq Z_t - Z_t \left(\frac{k - 4\Delta}{kn} \right)$$

Condition: Provided $k \geq 4\Delta + 1$, the expected distance strictly decreases:

$$\mathbb{E}[Z_{t+1}] \leq \mathbb{E}[Z_t] \left(1 - \frac{1}{kn} \right)$$

By induction, $\mathbb{E}[Z_t] \leq ne^{-t/kn}$. Using Markov's inequality, we can bound the probability $\Pr[Z_t > 0]$, showing the chains couple in $\leq kn(2 + \log n)$ steps. □

Recent Works and Improvements

The $k > 2\Delta$ bound by Jerrum (1995) [3] stood as a benchmark for Glauber dynamics, but significant effort has gone into lowering this threshold.

- **Vigoda (1999) [4]:** Introduced a new Markov chain that updates multiple vertices simultaneously, breaking the 2Δ barrier to achieve rapid mixing for $k > \frac{11}{6}\Delta$.
- **Delcourt et al. (2018) [5]:** Refined Vigoda's approach and coupling techniques to achieve $k \geq (\frac{11}{6} - \eta)\Delta$.
- **(2020–Present):** Recent breakthroughs using spectral bounding techniques have pushed the threshold closer to the theoretical limit of $k \geq \Delta + 2$ for certain classes of graphs.

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